SDC2 256-µstep speed characterization

Summary view: average speed and speed ratio computed from raw encoder count deltas over each 45 s segment.

Fig. 1 **Speed tracking summary at 256 microsteps**

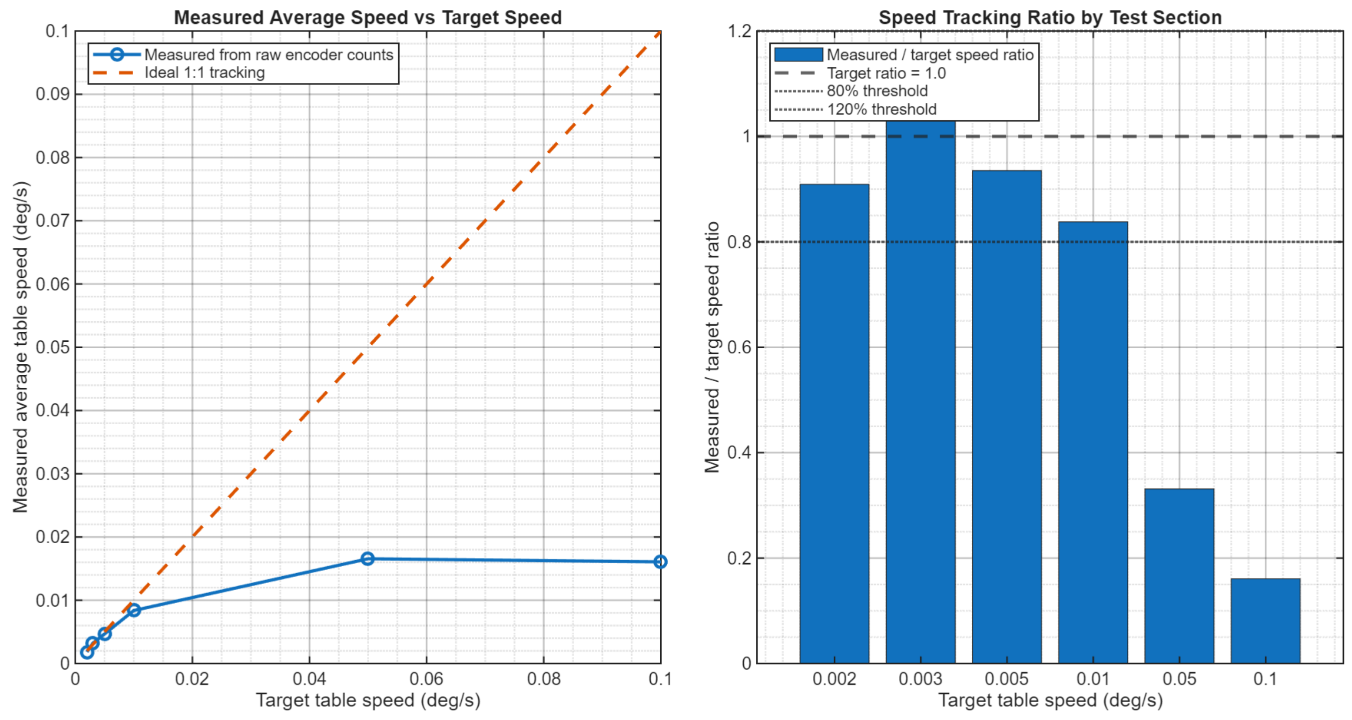

Figure 1. The low-speed region tracks reasonably from 0.002–0.010 deg/s. Higher requested speeds collapse to approximately the same measured speed ceiling.

Main message

SDC2 works as an ultra-slow motion platform at the current 256-µstep setting, but it is speed-limited above 0.010 deg/s.

BEST LOW-SPEED TRACKING

0.003 deg/s

Measured 0.003214 deg/s; ratio 1.071

UPPER USEFUL CASE

0.010 deg/s

Measured 0.008378 deg/s; ratio 0.838

SPEED CEILING

~0.016 deg/s

0.05 and 0.10 deg/s commands both plateau near this value.

NEXT TEST

128 / 64 / 32 µstep

Lower microstepping likely increases the usable higher-speed range.

Method note: raw encoder count events only; no filtering, smoothing, or grouped averaging.

Low-speed tracking is real and visible in the raw count traces

The count-delta plots are easiest to explain to the team: measured counts should follow the ideal dashed line.

Fig. 2A **Best success case: 0.003 deg/s**

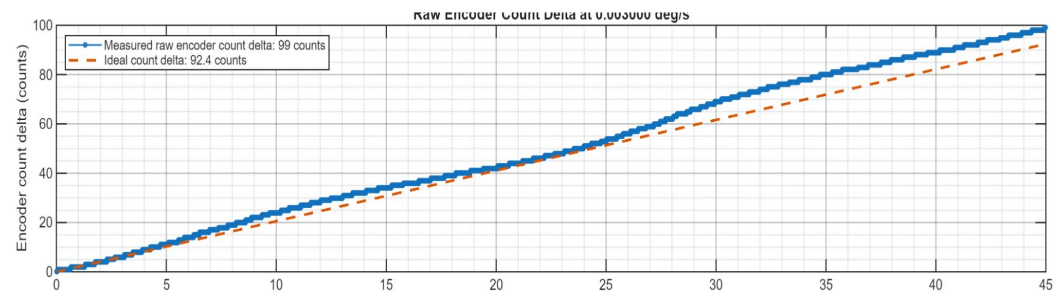

Figure 2A. 0.003 deg/s tracks very closely: measured 99 counts vs 92.4 ideal counts; average measured speed = 0.003214 deg/s; speed ratio = 1.071.

Fig. 2B **Upper useful case: 0.010 deg/s**

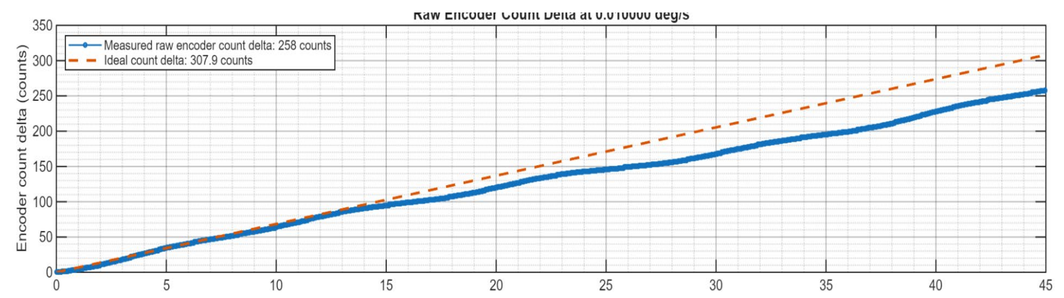

Figure 2B. 0.010 deg/s remains usable but under target: measured 258 counts vs 307.9 ideal counts; average measured speed = 0.008378 deg/s; speed ratio = 0.838.

Why these figures are shown

They show the raw encoder count trajectory directly. If the blue measured curve follows the orange ideal line, the table is physically moving at the commanded order of magnitude.

Average speed definition

Average measured speed = total raw encoder-count delta over the segment converted to table deg/s. This is the best single number for command-following accuracy.

RMS speed definition

RMS event speed = root-mean-square value of the raw event-to-event speed trace. Use it to compare speed ripple between microstep settings.

Suggested statement: “At 256 µstep, the cleanest low-speed operating region is about 0.003–0.005 deg/s; 0.010 deg/s is still usable but begins to show under-speed.”

Higher commanded speeds require a different microstep setting

At 256 μstep, higher commands do not create proportionally higher measured speeds.

Fig. 3A **0.050 deg/s command: actual motion falls far behind**

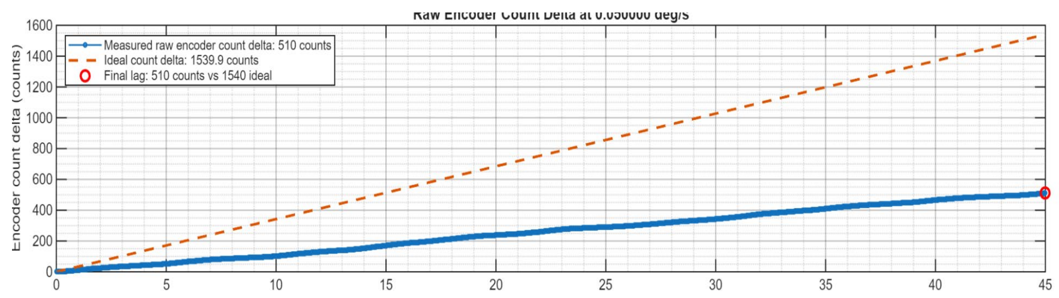

Figure 3A. 0.050 deg/s command: measured 510 counts vs 1539.9 ideal counts; average measured speed = 0.016560 deg/s; speed ratio = 0.331.

Fig. 3B **0.100 deg/s command: same speed ceiling**

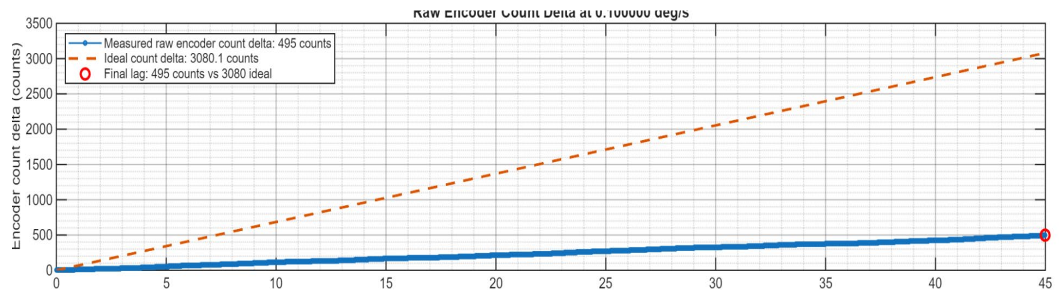

Figure 3B. 0.100 deg/s command: measured 495 counts vs 3080.1 ideal counts; average measured speed = 0.016071 deg/s; speed ratio = 0.161.

Key finding

The 0.050 and 0.100 deg/s commands both settle near ~0.016 deg/s actual table speed. That points to a real operating limit, not a plotting artifact.

Likely implication

256 μstep is strong for ultra-slow smoothness, but likely too fine for the higher-speed range. Lower microstep settings should raise the achievable command-following range.

Next comparison

- Repeat the same raw-event analysis at 128, 64, and 32 μstep
- Compare average speed ratio, RMS event speed, RMS speed error, and long/short gap counts
- Select the highest microstep setting that still tracks the required speed range

Definitions for interpreting the figures

These notes make the slide deck self-contained for team review and later documentation.

Encoder-event method

- Use raw EncoderCount changes only
- Event interval = time between consecutive encoder count changes
- Event speed = count step size / event interval
- No filtering, smoothing, or grouped averaging applied

Why this matters

The event-speed trace looks noisy because it is intentionally showing raw event-to-event timing behavior at the encoder resolution.

Gap definitions

expected interval = 1 / expected count rate

- Long gap: event interval > 2.0× expected interval
- Short gap: event interval < 0.5× expected interval
- Long gaps imply slow/stalled motion, slip, or delayed counts
- Short gaps imply catch-up motion or faster-than-expected count events

These are flags, not filtered results.

Metrics to add per run

- Average measured speed
- Measured / target speed ratio
- RMS event speed
- RMS speed error
- Long gap count
- Short gap count
- Final count lag

Use these metrics to compare 256, 128, 64, and 32 µstep settings on equal terms.

Recommended conclusion: current 256-µstep setup is valid for ultra-slow characterization, but faster SDC2 scans likely require lower microstepping plus a carefully limited feedback loop.